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Automation & Robotics

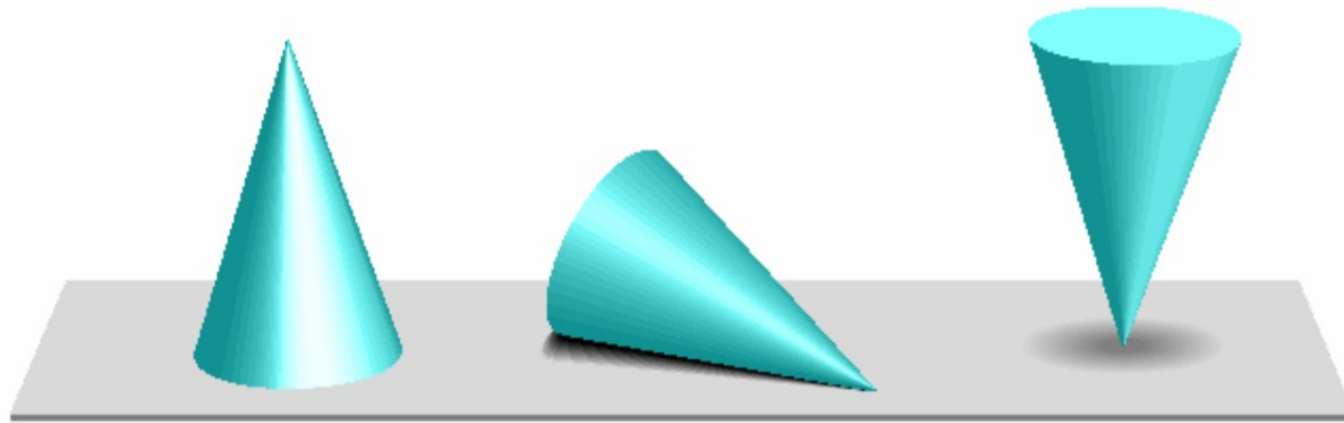
PID CONTROL

Routh-Hurwitz Stability Criterion

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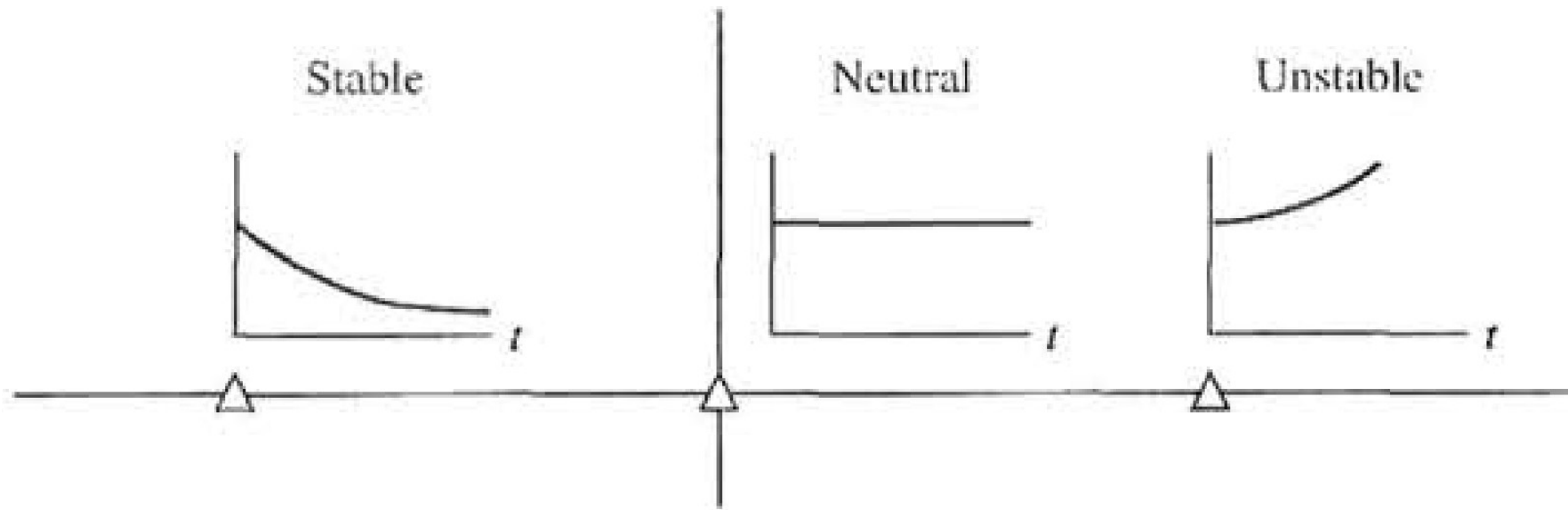
Routh-Hurwitz Criterion



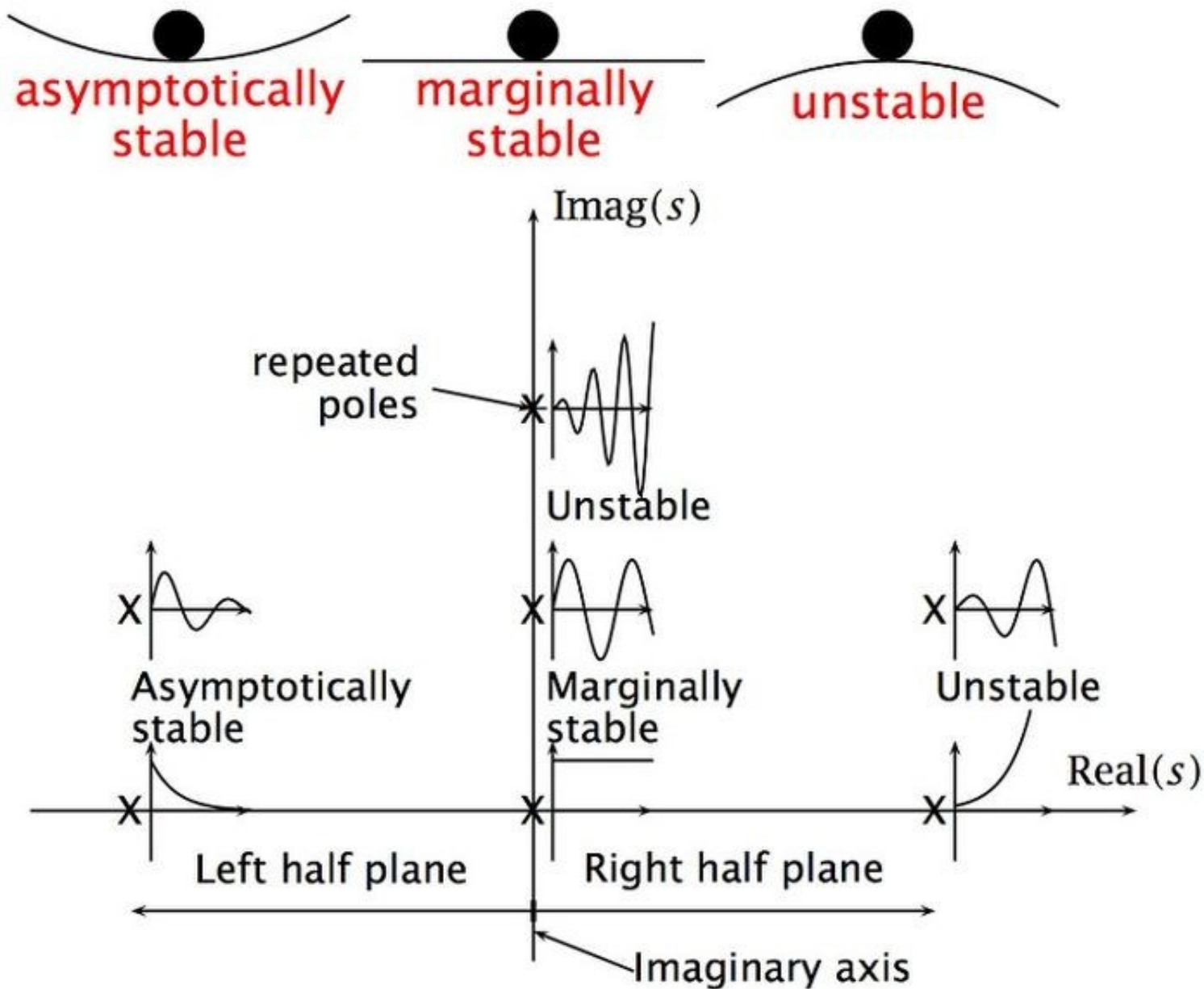
(a) Stable

(b) Neutral

(c) Unstable



Routh-Hurwitz Stability Criterion



Routh-Hurwitz Stability Criterion

The **Routh-Hurwitz criterion** is a necessary and sufficient criterion for the stability of linear systems. The method was originally developed in terms of determinants, but we shall use the more convenient array formulation.

The Routh-Hurwitz criterion is based on ordering the coefficients of the characteristic equation

$$a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \dots + a_1 s + a_0 = 0$$

Further rows of the schedule are then completed as where

$$\begin{array}{c|ccc} s^n & a_n & a_{n-2} & a_{n-4} \\ s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} \\ s^{n-2} & b_{n-1} & b_{n-3} & b_{n-5} \\ s^{n-3} & c_{n-1} & c_{n-3} & c_{n-5} \\ \vdots & \vdots & \vdots & \vdots \\ s^0 & h_{n-1} & & \end{array}$$

$$b_{n-1} = \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}} = \frac{-1}{a_{n-1}} \begin{vmatrix} a_n & a_{n-2} \\ a_{n-1} & a_{n-3} \end{vmatrix},$$

$$b_{n-3} = -\frac{1}{a_{n-1}} \begin{vmatrix} a_n & a_{n-4} \\ a_{n-1} & a_{n-5} \end{vmatrix},$$

$$c_{n-1} = \frac{-1}{b_{n-1}} \begin{vmatrix} a_{n-1} & a_{n-3} \\ b_{n-1} & b_{n-3} \end{vmatrix},$$

Routh-Hurwitz Stability Criterion

The Routh-Hurwitz criterion states that the number of roots of $q(s)$ with positive real parts is equal to the number of changes in sign of the first column of the Routh array.

Routh-Hurwitz Stability Criterion

Consider the characteristic polynomial

$$q(s) = s^5 + s^4 + 4s^3 + 24s^2 + 3s + 63.$$

The Routh array is

$$\begin{array}{c|ccc} s^5 & 1 & 4 & 3 \\ s^4 & 1 & 24 & 63 \\ s^3 & -20 & -60 & 0 \\ s^2 & 21 & 63 & 0 \\ s^1 & 0 & 0 & 0 \end{array}$$

Routh-Hurwitz Stability Criterion

$$q(s) = s^3 + 2s^2 + 4s + K,$$

where K is an adjustable loop gain. The Routh array is then

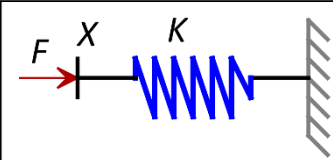
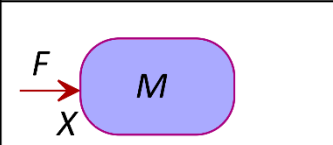
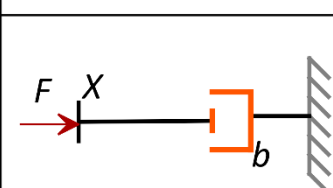
$$\begin{array}{c|cc} s^3 & 1 & 4 \\ s^2 & 2 & K \\ s^1 & \frac{8-K}{2} & 0 \\ s^0 & K & 0 \end{array}.$$

For a stable system, we require that

$$0 < K < 8.$$

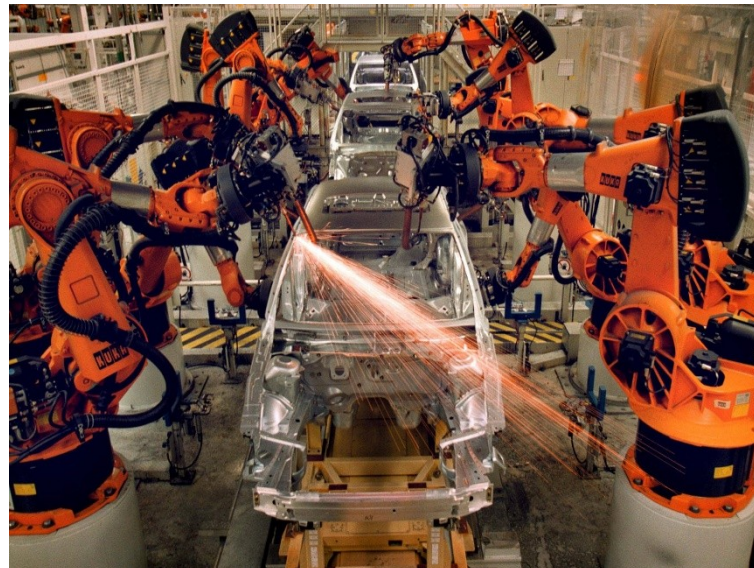
Recap on the previous lecture

- Mechanical interaction with objects/environment.
- Mechanical interaction dynamics can be characterised by mechanical impedance.

	Impedance $\frac{F}{\dot{X}}$	Admittance $\frac{\dot{X}}{F}$
	$\frac{K}{S}$	$\frac{S}{K}$
	MS	$\frac{1}{MS}$
	b	$\frac{1}{b}$

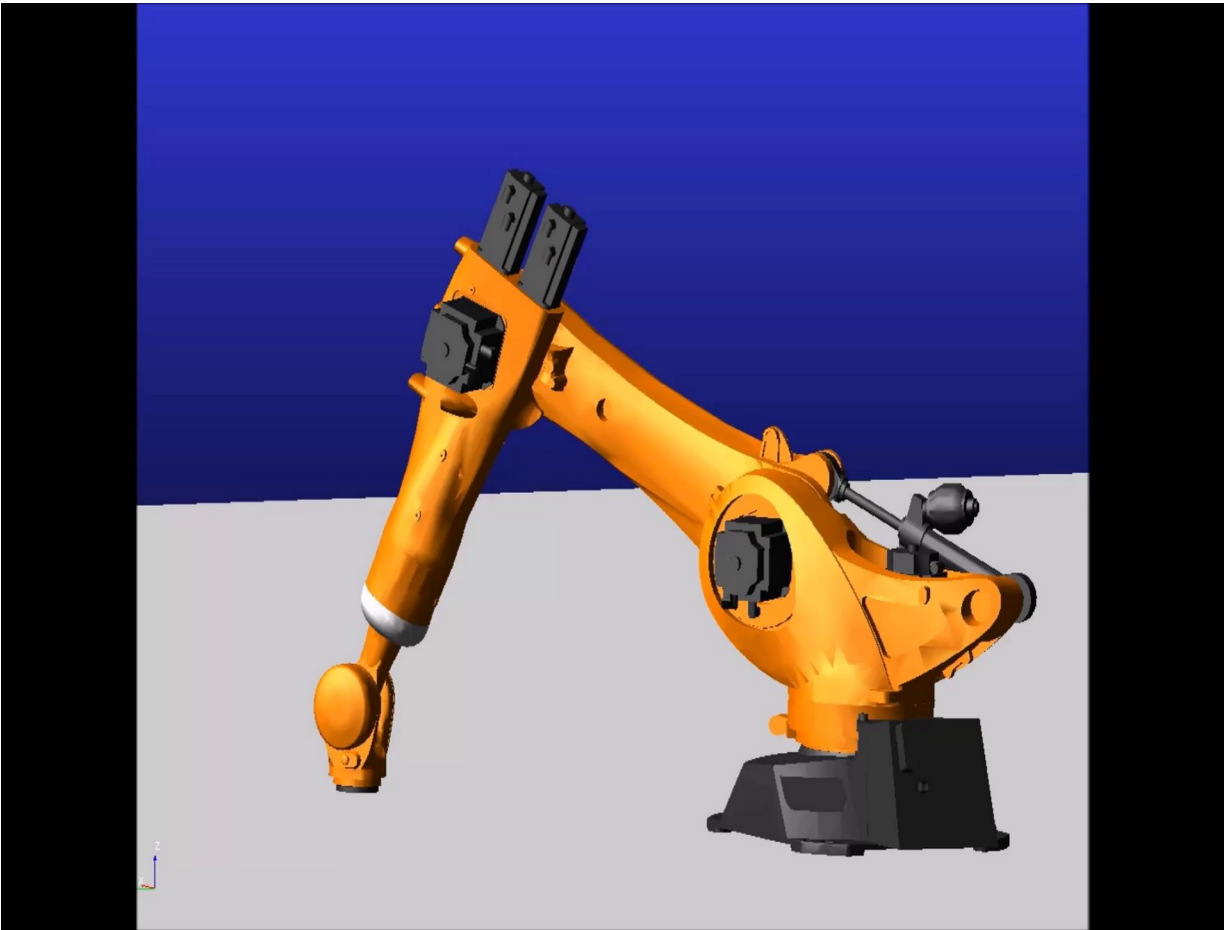
Laplace-Transformed Impedance and Admittance
Functions for Common Mechanical Elements

Examples of mechanical interactions



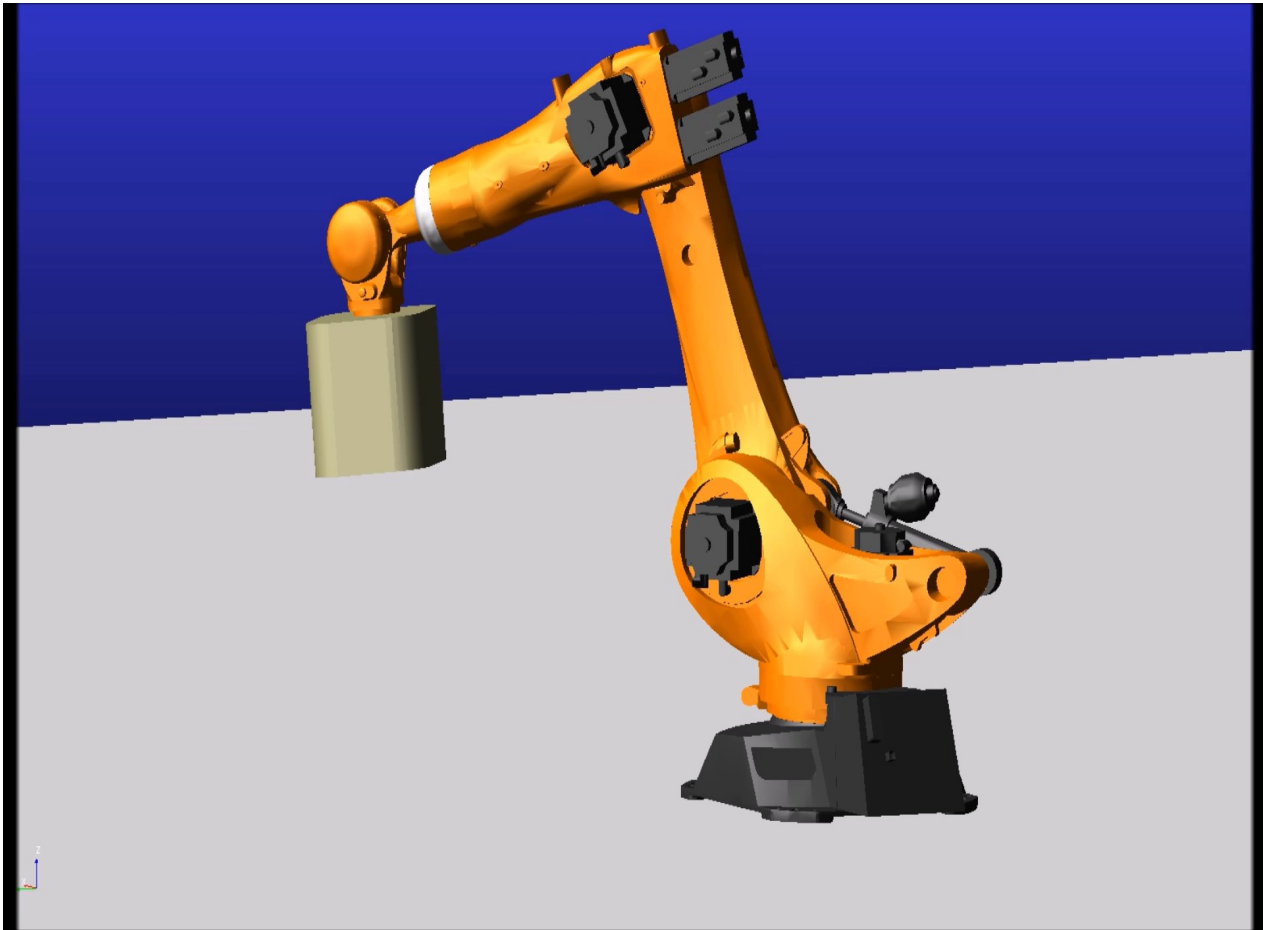
The Effect of Interaction on Performance and Stability

Robot in isolation



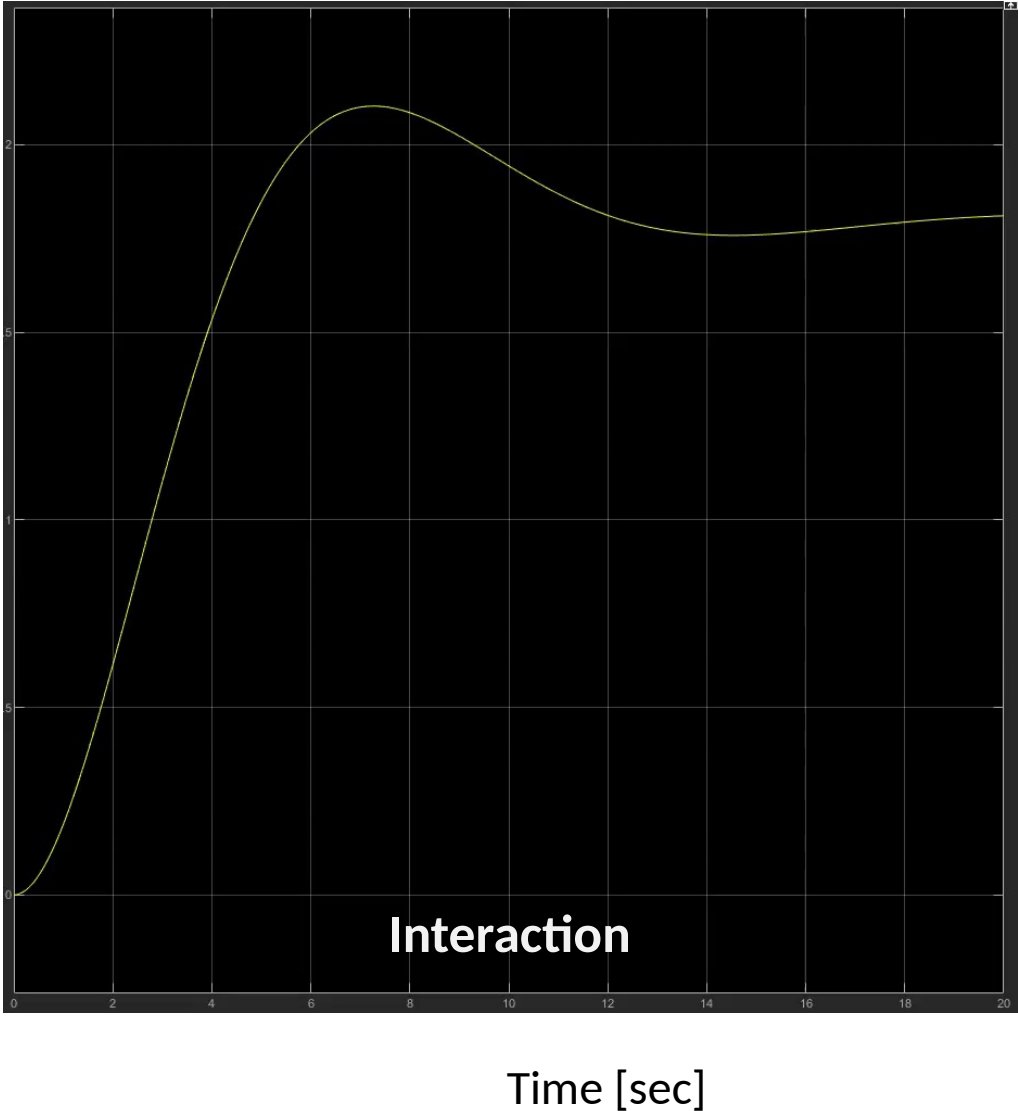
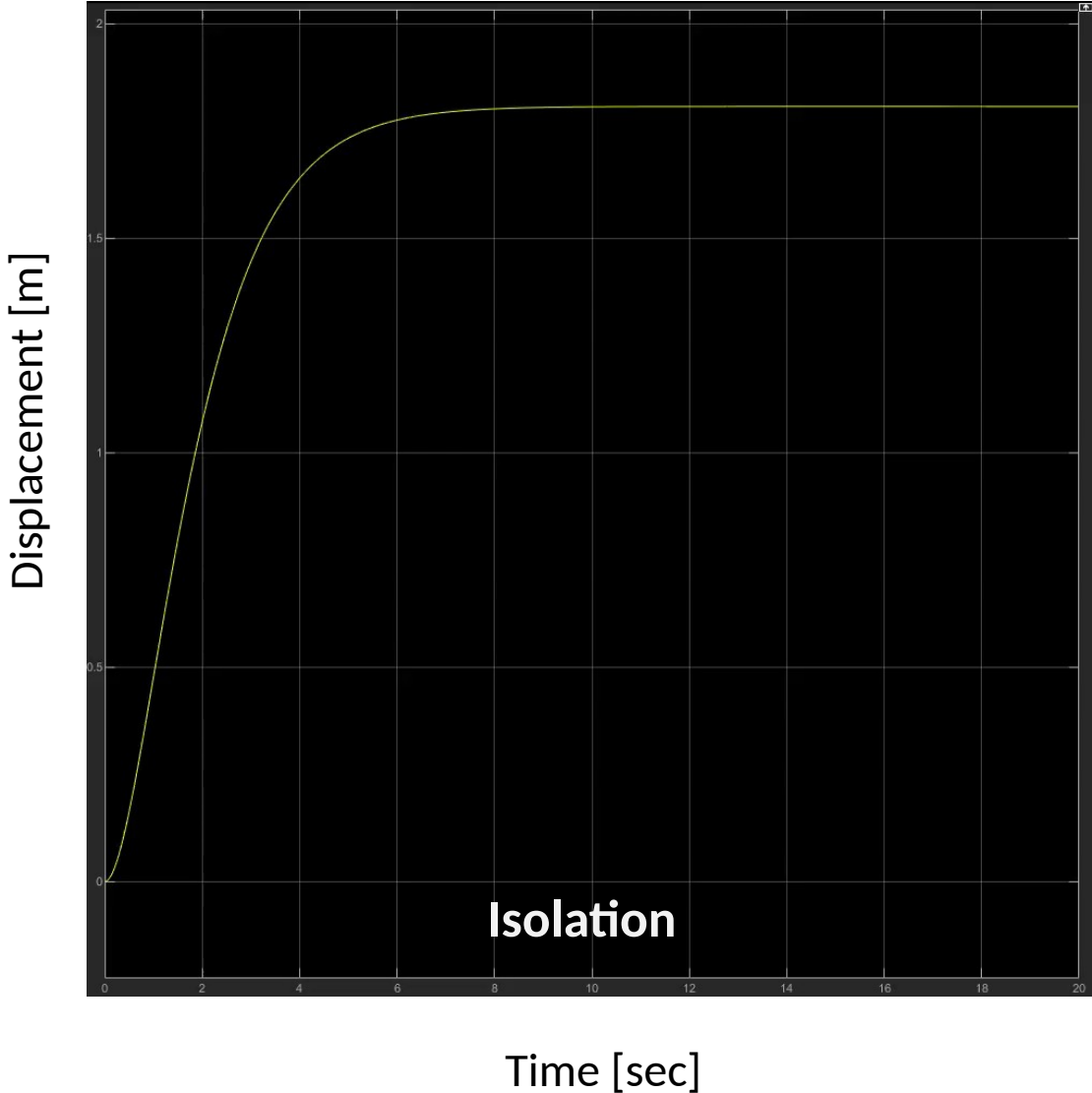
Robot body: Aluminium (2740 kg/m^3)
Robot total mass: 636 Kg
Kp, KI: 4.9, 1.2

Robot interacts with environment



Robot body: Aluminium (2740 kg/m^3)
Robot total mass: 636 Kg
Kp, KI: 4.9, 1.2
Object weight: 26.5 Kg

The Effect of Interaction on Performance and Stability



The Effect of Interaction on Performance and Stability

When interaction occurs

- The dynamic properties of the environment are important.
- It can cause error on force or motion control.
- Stability may also be compromised.
- A system that is stable in isolation can become unstable (coupled or contact instabilities)

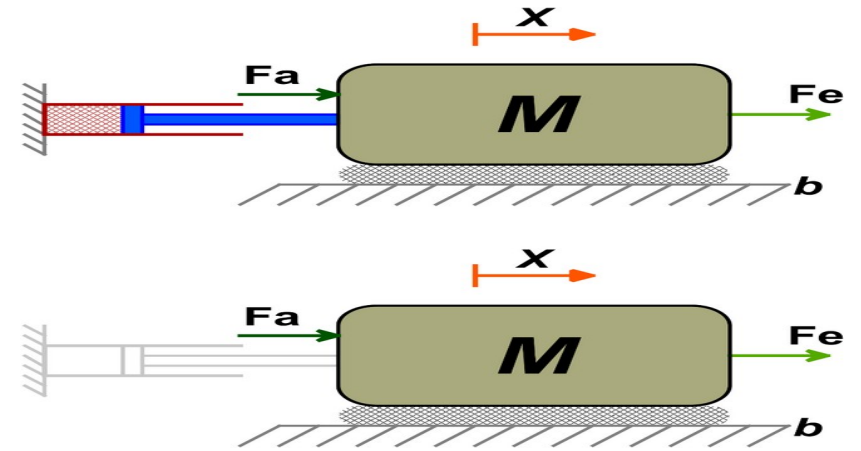
Example1: Robot motion controller

M : represents the robot's inertial properties

F_a : actuator forces

F_e : environmental forces

b : a damper represents frictional losses



Simplified model of a robot with inertial, frictional, actuator, and environmental forces.

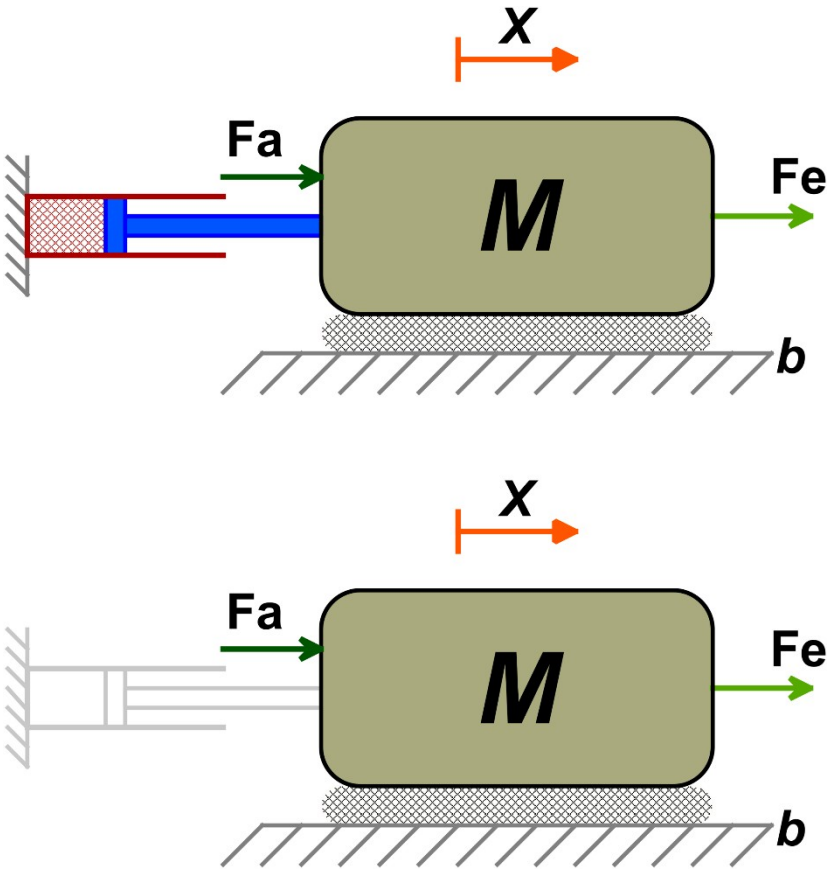
Example1: Robot motion controller

Newton's laws of motion

(1)

The Laplace-transformed equation of motion

$X =$ (2)



Example1: Robot motion controller

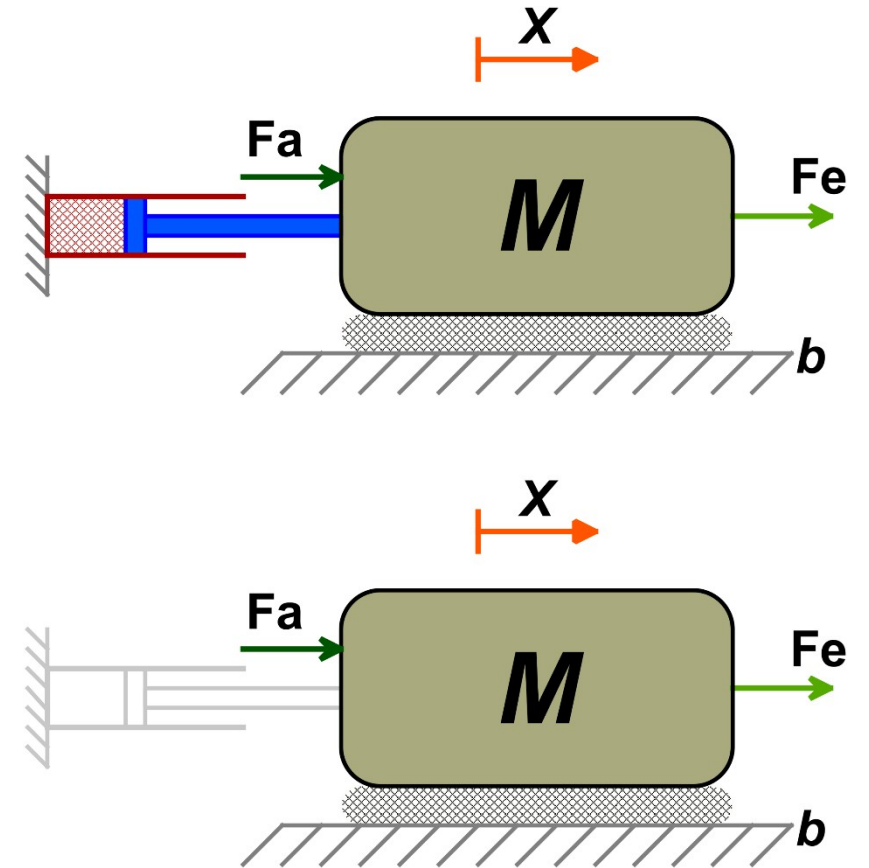
A proportional-integral motion controller is applied

(3)

R: Laplace transform of the reference position

K_p: Proportional gain

K_i: Integral gain

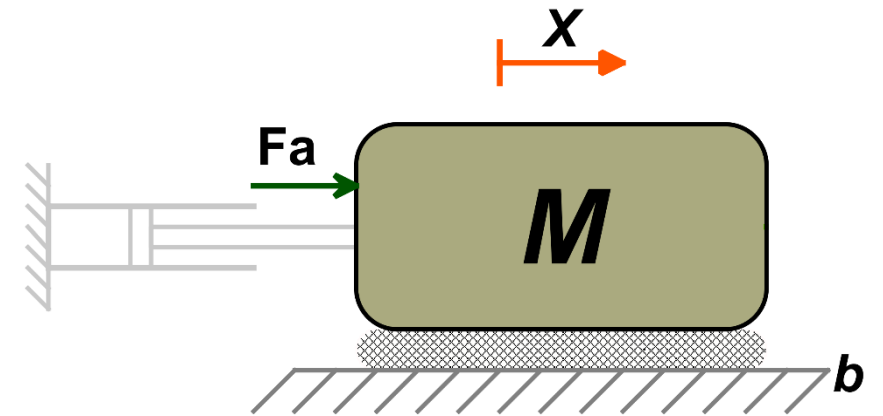


Robot in isolation

In isolation **$F_e = 0$**

The closed-loop transfer function is

(4)



Simplified model of a robot with inertial, frictional, actuator, and environmental forces in isolation

From the Routh-Hurwitz stability criterion

$$K_I < \frac{b K_P}{M} \quad (5)$$

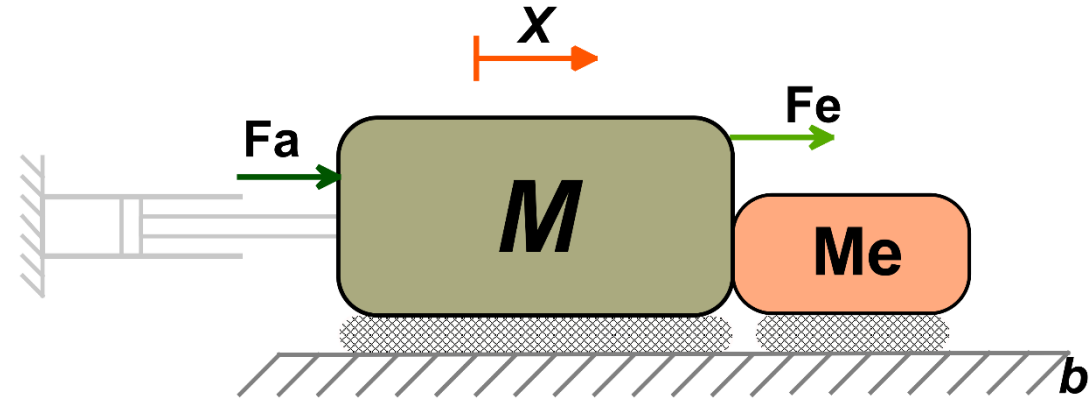
A condition for isolated stability

Robot interacts with environment

However, this condition is not sufficient to ensure that the robot will remain stable when it interacts with objects in its environment. Even the simple act of grasping an object may be destabilizing. If the system is not isolated, and instead is coupled to a mass M_e , this is equivalent to increasing the mass from M to $(M + M_e)$. Hence a condition for stability of the coupled system is

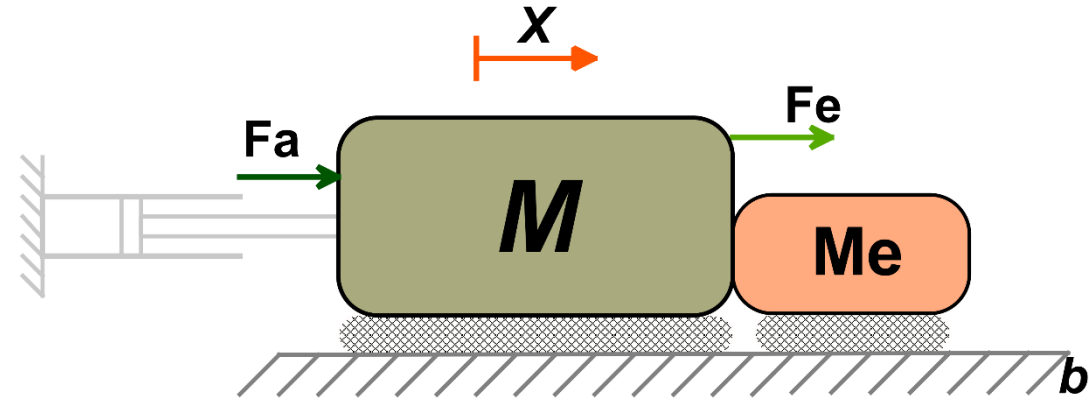
$$K_I < \frac{b K_P}{(M + M_e)}$$

(6)



Robot interacts with environment

However, for any fixed controller gains, coupling the robot to a sufficiently large m_e will violate the condition for stability. Interaction with a large enough mass can always destabilize a system that includes integral-action motion control, even if that system is stable in isolation



$$K_I < \frac{b K_P}{(M + M_e)}$$

(6)

Robot interacts with environment

This difficulty may be obscured by the facts that, at present, most robots have inertia far exceeding that of the payloads they may carry, the total mass $M + M_e$ is not much greater than the mass M of the robot alone, and the bounds of Equation (5) and Equation (6) are similar.

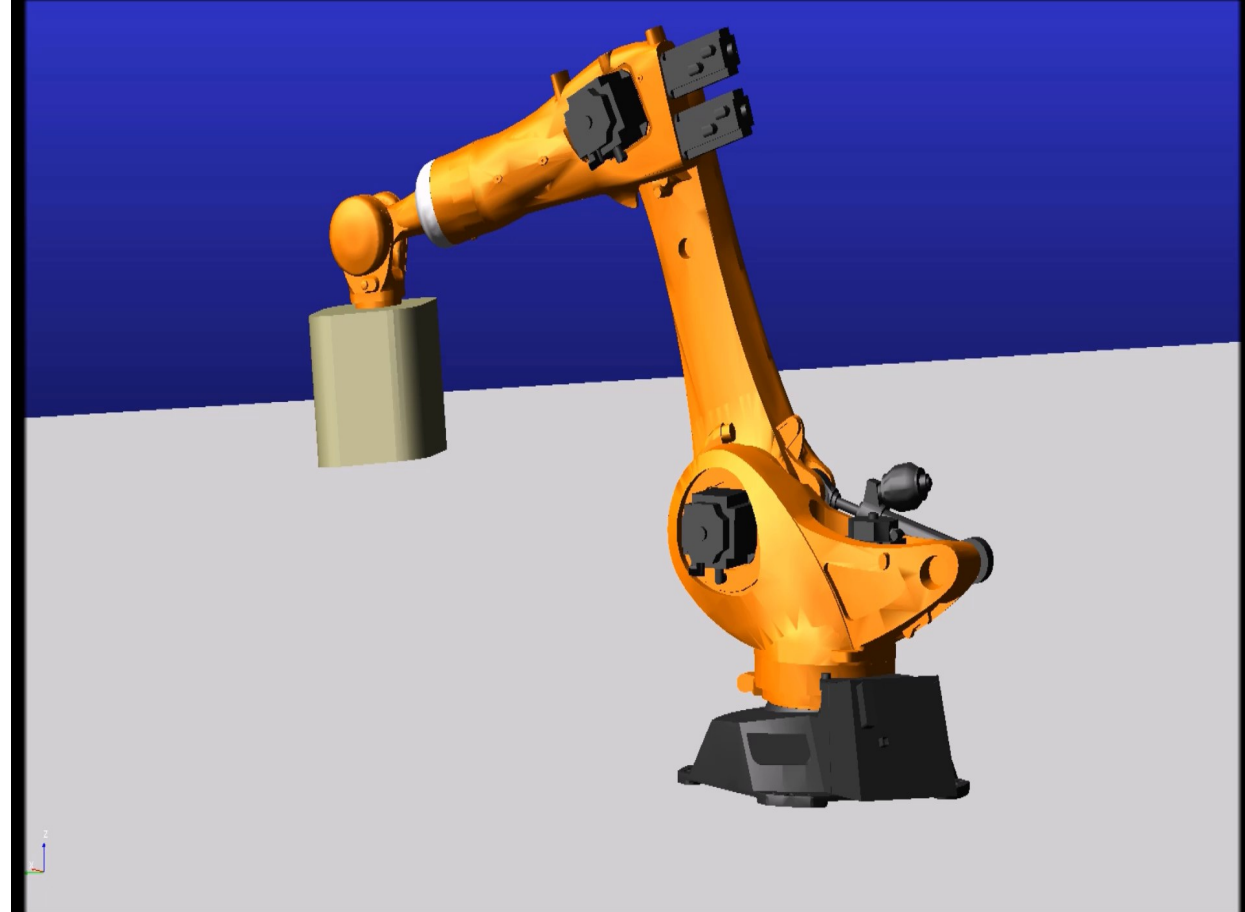
$$K_I < \frac{b K_P}{(M)} \quad (5)$$

$$K_I < \frac{b K_P}{(M + M_e)} \quad (6)$$

Example



Robot body:	Cast Iron (7080 kg/m^3)
Robot total mass:	1643 Kg
KI:	1.2
Object weight:	26.5 Kg



Robot body:	Aluminium (2740 kg/m^3)
Robot total mass:	636 Kg
KI:	1.2
Object weight:	26.5 Kg

Example

